

Advanced (Habanero)

- Q1.** What is the slope of a line parallel to $y = 2x + 3$?
- (A) $m = 1$
(B) $m = 2$
(C) $m = 3$
(D) $m = 4$
(E) $m = 5$
- Q2.** Which equation represents a line parallel to $y = -3x - 2$?
- (A) $y = -3x + 1$
(B) $y = -2x - 3$
(C) $y = 3x - 2$
(D) $y = 2x + 1$
(E) $y = -4x - 1$
- Q3.** If $y = mx + b$ is parallel to $y = 2x - 1$, what is m ?
- (A) $m = 1$
(B) $m = 2$
(C) $m = 3$
(D) $m = 4$
(E) $m = 5$
- Q4.** Which of the following lines is parallel to $y = \frac{1}{2}x + 2$?
- (A) $y = \frac{1}{3}x - 1$
(B) $y = \frac{1}{2}x - 3$
(C) $y = 2x + 1$
(D) $y = \frac{2}{3}x + 4$
(E) $y = -\frac{1}{2}x + 1$
- Q5.** What is the equation of a line parallel to $y = -2x + 1$ and passing through $(1, 3)$?
- (A) $y = -2x + 5$
(B) $y = -2x + 3$
(C) $y = 2x - 1$
(D) $y = -2x + 1$
(E) $y = -3x + 2$
- Q6.** If two lines are parallel, what is the relationship between their slopes?
- (A) They are equal
(B) They are perpendicular
(C) They are opposite reciprocals
(D) They are unequal
(E) They are negative reciprocals
- Q7.** Which of the following statements is true about parallel lines?
- (A) They intersect at exactly one point
(B) They are perpendicular to each other
(C) They have the same slope
(D) They have different equations
(E) They are not defined
- Q8.** What is the slope of a line parallel to $y = \frac{3}{4}x - 2$?
- (A) $m = \frac{1}{4}$
(B) $m = \frac{3}{4}$
(C) $m = \frac{2}{3}$
(D) $m = \frac{4}{3}$
(E) $m = -\frac{3}{4}$

Open Ended Questions (FRQ)

- Q9.** Write the equation of a line parallel to $y = 2x - 1$ and passing through $(2, 5)$. Show all work and explain your reasoning.
- Q10.** Prove that the lines $y = 3x + 2$ and $y = 3x - 4$ are parallel. Use the definition of parallel lines and show all work.

Chef's Tips

- Use a graphing calculator to check your work
- Make sure to show all work and explain your reasoning
- Check for any special cases or exceptions